

Stage 1 Portfolio choice

Introduction

Portfolio universe and data

We use 29 stocks from the Dow Jones Industrial Average roster dated 31 October 2015, retaining members that later leave the index.^[1] The sample contains 130 monthly USD returns from November 2015 to August 2026, the latest complete month also covered by the official factor release. Instructor prices take priority for covered ordinary months;^[2] Yahoo Finance supplies other stocks and date extensions, with dividend and stock-split adjustments.^[3] We account for mandatory spin-off distributions in the return series using a consistent reinvestment convention. Old E. I. du Pont is excluded because identity-consistent 2015–2017 historical prices could not be recovered from the tested sources; the current DD series could not substitute for the missing old-DuPont history. October 2015 is the return baseline. UTX continues under RTX. Monthly risk-free returns are the Kenneth French one-month Treasury-bill series, converted from published percentages to decimals.^[4]

Later index removals

2015 member	Index removal date	Later return series
GE	2018-06-26 ^[5]	GE retained
XOM	2020-08-31 ^[6]	XOM retained
PFE	2020-08-31 ^[6]	PFE retained
UTX (later RTX)	2020-08-31 ^[6]	RTX continuation
INTC	2024-11-08 ^[7]	INTC retained
VZ	2026-06-29 ^[8]	VZ retained

We keep these stocks after index removal and do not add their index replacements. This historical membership rule helps mitigate potential upward bias in historical returns from selecting only current constituents.

1a Summary statistics

We estimate arithmetic means and the sample covariance of raw monthly returns. Here $\mathbf{R}_t$ is the 29-stock monthly return vector, and the full-sample estimates are

$$\hat{\mu} = \frac{1}{130} \sum_{t=1}^{130} \mathbf{R}_t,$$

$$\hat{\Sigma} = \frac{1}{129} \sum_{t=1}^{130} (\mathbf{R}_t - \hat{\mu})(\mathbf{R}_t - \hat{\mu})^T.$$

Monthly means and volatility are shown as percentages. Blue shading identifies retained DJIA members; amber identifies later index removals. Calculations retain full precision; displayed values round to three decimals.

Exhibit 1a Monthly return statistics for all 29 stocks, each with 130 common observations.

Stock	Mean % (monthly)	Volatility % (monthly)	Stock	Mean % (monthly)	Volatility % (monthly)
MMM	0.703	6.887	JPM	1.782	6.876
AXP	1.566	7.517	MCD	0.978	4.846
AAPL	2.226	7.917	MRK	1.240	6.002
BA	0.977	11.099	MSFT	2.081	6.556
CAT	2.465	9.210	NKE	0.040	8.019
CVX	1.305	8.001	PFE	0.490	6.486
CSCO	1.533	7.120	PG	0.837	4.759
KO	0.937	4.613	TRV	1.250	5.710
XOM	1.183	7.940	RTX	1.512	7.496
GE	1.314	10.344	UNH	1.384	8.085
GS	1.817	8.239	VZ	0.637	5.447
HD	1.160	6.478	V	1.444	5.617
INTC	1.831	14.775	WMT	1.618	5.398
IBM	1.090	7.909	DIS	0.380	8.530
JNJ	1.095	4.844			

1b Portfolio weights

Each window uses its own average risk-free return. The unrestricted, fully invested tangency portfolio is

$$z = \hat{\Sigma}^{-1}(\hat{\mu} - \overline{R}_F \mathbf{1}), \quad w_F = \frac{z}{\mathbf{1}^T z}.$$

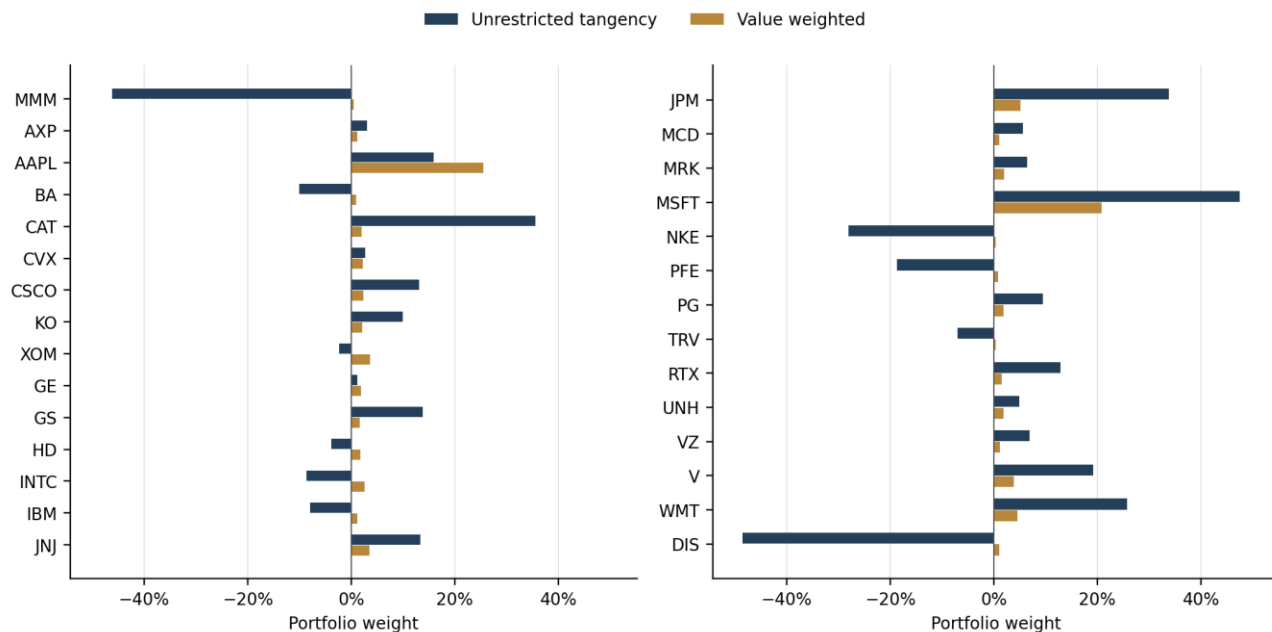
All estimated covariance matrices are positive definite and normalization denominators are positive; no shrinkage, pseudoinverse or clipping is used.

Value weights normalize Yahoo company market capitalizations dated 31 August 2026 over the same 29 stocks.^[3] We hold this sample-end vector fixed and rebalance each portfolio to its specified weights monthly. It is a descriptive comparator. Model Sharpe uses annualized arithmetic mean excess return and raw-return volatility. Monthly portfolio and risk-free means are multiplied by 12, and volatility by the square root of 12:

$$S_{model} = \frac{\hat{\mu}_{p,ann} - \hat{\mu}_{RF,ann}}{\hat{\sigma}_{p,ann}}.$$

We compare three portfolios over the same 29 stocks. Unrestricted tangency uses the full-sample estimates, with weights summing to one and short sales permitted. Value weighting allocates in proportion to company market capitalization. Long-only tangency maximizes the same model Sharpe with nonnegative weights summing to one, testing the effect of prohibiting short sales.

Exhibit 1b Tangency and value weights for all 29 stocks share a common scale. Negative weights denote short sales; the unrestricted optimum contains large offsetting positions.



Portfolio	Mean %	Volatility %	Model Sharpe	Gross %
Unrestricted tangency	44.547	19.794	2.142	462.588
Value weighted	20.790	15.712	1.186	100.000
Long only	22.207	13.946	1.438	100.000

Tangency and value weighting produce markedly different portfolio weights. MSFT receives 47.520% in tangency versus 20.785% by size; DIS has a -48.498% tangency weight versus 1.025% by size. Long positions total 281.294% and shorts 181.294%, giving 462.588% gross exposure despite 100% net investment.

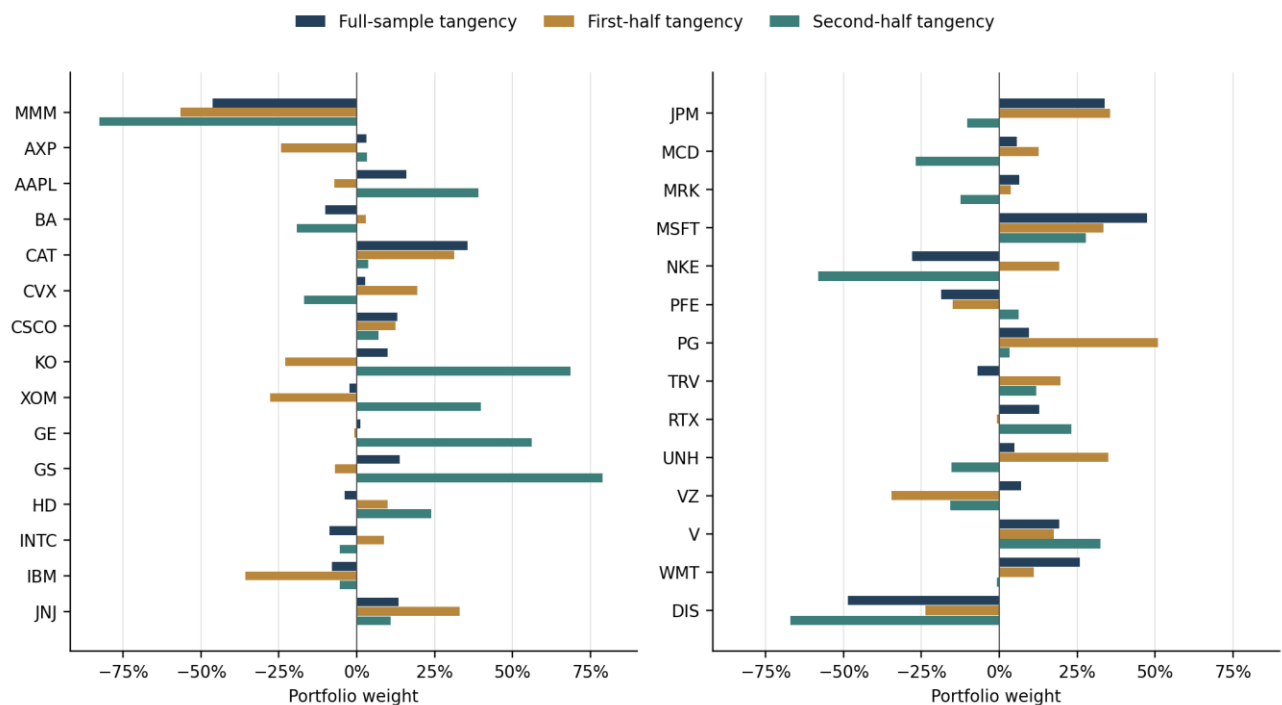
In sample, unrestricted tangency has an annualized arithmetic mean of 44.547%, volatility of 19.794% and model Sharpe of 2.142. The corresponding figures are 20.790%, 15.712% and 1.186 for value weighting, and 22.207%, 13.946% and 1.438 for long-only tangency. Prohibiting short sales lowers the fitted maximum Sharpe because nonnegative weights narrow the feasible set and remove leveraged long-short allocations.

Requiring nonnegative weights gives 9 positive positions, a largest weight of 28.609% in MSFT, 100% gross exposure and model Sharpe 1.438. This removes extreme offsetting holdings at the cost of a lower fitted optimum, without guaranteeing better subsequent returns.

Value weighting allocates by company size; tangency weighting uses estimated risk premia and covariance. Inverse covariance can amplify errors along low-variance directions, producing large offsetting positions in correlated assets. Maximizing Sharpe on the evaluation sample favors the fitted portfolio, so its higher ratio does not establish superior future performance.

1c Stability and interpretation

Exhibit 1c Full-sample weights provide a baseline for the two half-sample estimates. Between the halves, 17 of 29 holdings change sign and the weight vectors correlate 0.106.



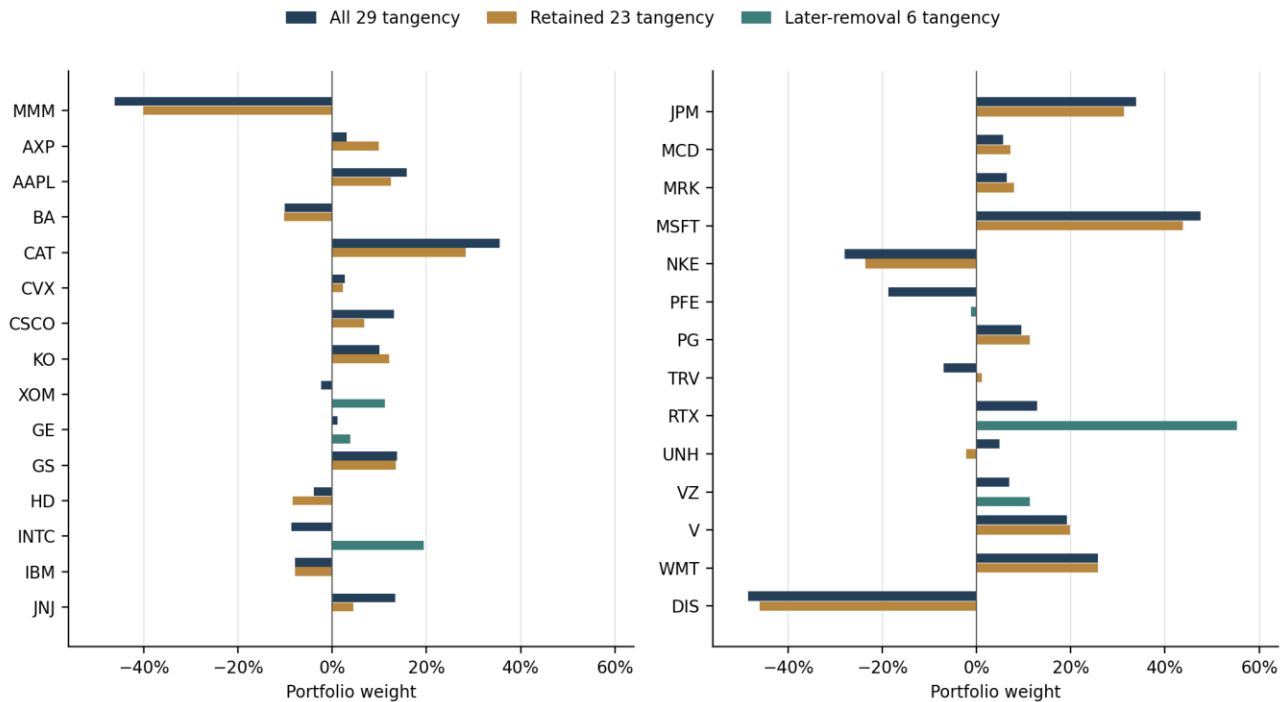
Portfolio	Mean %	Volatility %	Model Sharpe	Gross %
Full-sample tangency	44.547	19.794	2.142	462.588
First-half tangency	44.215	13.739	3.146	612.580
Second-half tangency	84.477	23.051	3.521	772.765

The full sample is November 2015 to August 2026 (130 months); the halves are November 2015 to March 2021 and April 2021 to August 2026 (65 months each). Each portfolio independently estimates means, covariance and average RF in its own window. The table reports fitted performance in each estimation window; mean and volatility are annualized percentages, and gross exposure is the sum of absolute weights. KO moves from -22.893% to 68.511% between the halves. Separately, applying first-half fitted weights to second-half returns reduces model Sharpe from 3.146 to 0.432. Sampling error and changing underlying parameters can both contribute; this split does not identify either explanation alone.

Universe sensitivity

We compare the primary 29-stock cohort with a 23-stock subset excluding GE, XOM, PFE, RTX, INTC and VZ, the six later DJIA removals disclosed above. Both unrestricted tangency portfolios are re-estimated and evaluated over November 2015 to August 2026 (130 months), using the same return data and average RF. This tests sensitivity to universe selection. A separate six-stock portfolio is added below to compare the two disjoint groups directly.

Universe comparison Weights for all 29 stocks, the retained 23 and the six later removals share a common scale. Stocks outside a subset have zero weight because they are not held.



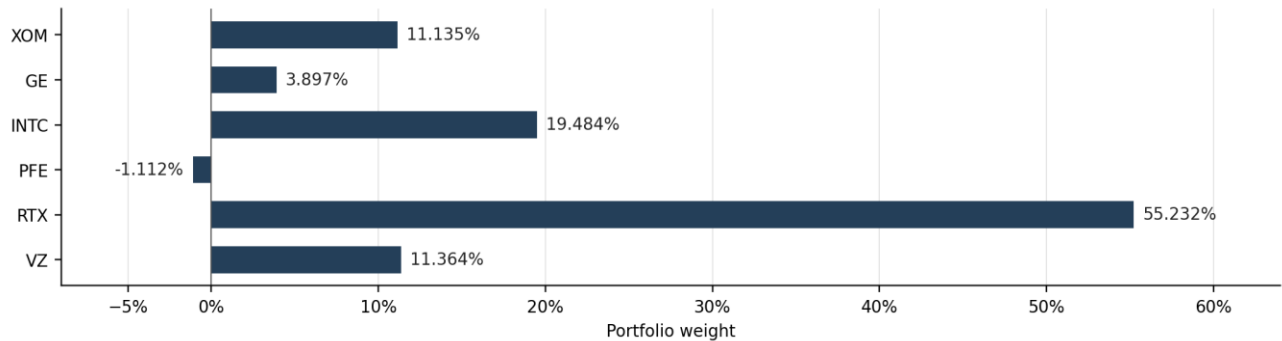
Portfolio	Mean %	Volatility %	Model Sharpe	Gross %
All 29 tangency	44.547	19.794	2.142	462.588
Retained 23 tangency	40.144	18.350	2.071	376.473
Later-removal 6 tangency	17.298	20.907	0.725	102.224

The table reports fitted performance over the same 130 months; mean and volatility are annualized percentages. Across the 23 common stocks, weights correlate 0.983 and only TRV and UNH change sign. JNJ has the largest absolute change, falling by 8.809 percentage points. Excluding the six later removals lowers model Sharpe from 2.142 to 2.071 and gross exposure from 462.588% to 376.473%.

Six later index removals

To compare the fitted risk-return performance of later-removed members with that of retained members, we estimate an independent tangency portfolio using only GE, XOM, PFE, RTX, INTC and VZ. These six stocks and the retained 23 form disjoint groups. The six-stock weights are fitted from their own mean vector and covariance matrix over the same 130 months, rather than obtained by rescaling their weights in the 29-stock portfolio. The six stocks left the DJIA; index removal does not itself imply bankruptcy or exchange delisting.

Six-stock weights The independent tangency portfolio is concentrated in RTX; negative weights denote short sales.



The six-stock portfolio has an annualized mean of 17.298%, volatility of 20.907% and model Sharpe of 0.725, compared with 40.144%, 18.350% and 2.071 for the retained 23. Gross exposure is 102.224% versus 376.473%. Within the six-stock portfolio, RTX receives 55.232% and PFE is short 1.112%, while net investment remains 100%. The direct subset comparison therefore shows a clearer difference in fitted performance than the overlapping 29-versus-23 comparison.

Both subsets are selected using later index status and are ex-post comparators. The 29-stock feasible set contains both subset feasible sets by assigning zero weights to omitted stocks, so its optimized in-sample Sharpe cannot be lower. The retained 23 and later-removal six also offer different numbers and types of investment opportunities: their fitted Sharpe difference reflects composition, covariance and optimization, rather than a causal effect of index removal.

Conclusion

The estimated tangency portfolio has a higher in-sample model Sharpe than value weighting (2.142 versus 1.186). Historical membership retains six later index removals; subsample estimation and a no-short-sale constraint reveal allocation sensitivity. The same-window universe comparison retains a 0.983 correlation across common-stock weights. Separately, the six later removals have a fitted model Sharpe of 0.725 versus 2.071 for the retained 23.

References

- [1] SEC / SPDR DIA Annual Report October 31 2015, Schedule of Investments, printed p.2
- [2] FE5108 instructor price bundle, prices_us_monthly.csv; original retained unchanged
- [3] Yahoo Finance cached daily chart data and native monthlyMarketCap dated August 31 2026
- [4] Kenneth French Data Library, US Fama French research factors, 202608 release
- [5] S&P Dow Jones Indices, Walgreens Boots Alliance set to join, June 19 2018
- [6] S&P Dow Jones Indices, Salesforce Amgen and Honeywell set to join, August 24 2020
- [7] S&P Dow Jones Indices, NVIDIA and Sherwin Williams set to join, November 1 2024
- [8] S&P Dow Jones Indices, Alphabet set to join, June 23 2026